

# (Implicit) Ensembles of Ensembles: Epistemic Uncertainty Collapse in Large Models

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# Motivation: Epistemic Uncertainty in Decision-Making

**Aleatoric** uncertainty reflects irreducible noise in the data. **Epistemic** uncertainty reflects the limits of a model's knowledge and is, in principle, reducible with additional data.

Reliable epistemic uncertainty is required in several downstream tasks:

- **Active learning and Bayesian experimental design:** selecting inputs whose acquisition would most reduce epistemic uncertainty.
- **Out-of-distribution detection:** identifying inputs that are unlike the training distribution.
- **Safety-critical deployment:** abstaining when the model's internal confidence is insufficient.

## Main question

As model capacity increases, can standard disagreement-based uncertainty estimates underestimate epistemic uncertainty?

# Overview of Contributions

- 1 Identifies *epistemic uncertainty collapse*: a decrease in disagreement-based epistemic uncertainty as model or ensemble size grows.
- 2 Provides a theoretical analysis for explicit *ensembles of ensembles*.
- 3 Proposes *implicit ensembling* as a possible explanation for similar collapse in wide neural networks.
- 4 Introduces *implicit ensemble extraction* to recover hidden diversity from a single trained model.
- 5 Evaluates the phenomenon across toy regression, MLPs, ResNet-style ensembles, and pre-trained vision models.

# Background: Bayesian Model Average

Let  $p(\theta)$  be the distribution over parameters you obtain by re-training with different random seeds. For a new input  $x^*$ :

$$p(y^* | x^*) = \int p(y^* | x^*, \theta) p(\theta) d\theta \quad (\text{BMA})$$

**Deep ensembles** approximate this by Monte-Carlo:

$$p(Y | x) \approx \frac{1}{M} \sum_{j=1}^M p(Y | x, \theta_j), \quad \theta_j \sim p(\theta).$$

*Caveat.* Deep ensembles are not exact Bayesian inference, but their diversity across loss-landscape modes captures a lot of what a true posterior would see (Wilson & Izmailov 2020; Fort, Hu, Lakshminarayanan 2019).

# Background: Decomposing Total Uncertainty

The Smith & Gal (2018) decomposition splits total uncertainty into two parts:

$$H[Y | x] = - \int p(Y | x) \log p(Y | x) dY \quad \text{total uncertainty}$$

$$\mathbb{I}[Y; \Theta | x] = H[Y | x] - \mathbb{E}_{p(\theta)}[H[Y | x, \theta]] \quad \text{epistemic}$$

**Reading of MI.** “If I knew  $\theta$ , by how much would the entropy of  $Y$  drop?”

- Large  $\Rightarrow$  parameters carry unused knowledge  $\Rightarrow$  *high* epistemic uncertainty.
- Zero  $\Rightarrow$  knowing  $\theta$  tells us nothing new  $\Rightarrow$  *no* epistemic uncertainty.

# Background: Estimating MI from an Ensemble

For an ensemble of  $M$  independently trained models,

$$\mathbb{I}[Y; \Theta \mid x] \approx \underbrace{H\left(\frac{1}{M} \sum_j p(Y \mid x, \theta_j)\right)}_{\text{entropy of mean}} - \underbrace{\frac{1}{M} \sum_j H(p(Y \mid x, \theta_j))}_{\text{mean of entropies}}.$$

**Key observation.** This is a *disagreement* score.

- All members predict the same distribution  $\Rightarrow$  both terms are equal  $\Rightarrow$  MI = 0.
- Members disagree strongly  $\Rightarrow$  mean entropy is much larger than mean of entropies  $\Rightarrow$  MI is large.

**Every “collapse” claim in the paper is ultimately:** *the members no longer disagree.*

# Uncertainty Reduction vs. Uncertainty Collapse

Lower epistemic uncertainty is not always bad.

## Reasonable reduction

- A larger model fits the true function better.
- Uncertainty decreases mainly near data-supported regions.
- OoD or data-sparse regions remain uncertain.

## Collapse

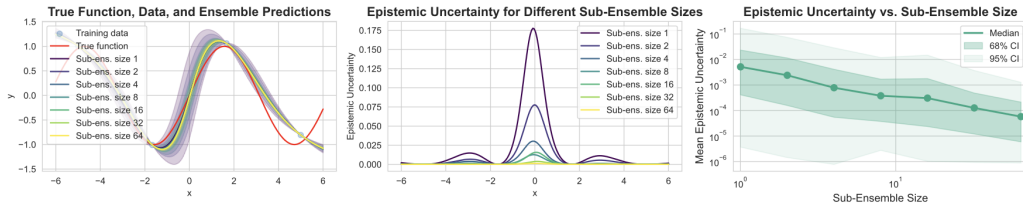
- Uncertainty decreases because averaging hides disagreement.
- MI shrinks even when accuracy is nearly unchanged.
- OoD or data-sparse regions also become overconfident.

## Key distinction

The paper focuses on uncertainty reduction caused by averaging away disagreement, rather than by genuine improvement in learned knowledge.

# A First Illustration of Uncertainty Collapse

Ground truth  $f(x) = \sin(x) + \epsilon$ , 4 training points;  $S = 10$  sub-ensembles of varying size  $M \in \{1, 2, 4, 8, 16, 32, 64\}$  (1-hidden-layer MLP, tanh).



## What to notice:

- Predictions converge to the true function (good).
- **Between training points**, where the model really doesn't know, the epistemic uncertainty band *shrinks by decades* as  $M$  grows.
- This is not desirable behaviour for active learning or OoD.



# Theory: Ensembles of Ensembles, formally

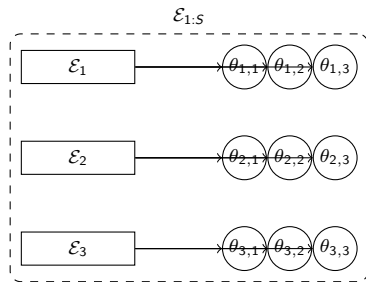
## Structure

- $S$  sub-ensembles  $\mathcal{E}_1, \dots, \mathcal{E}_S$ .
- Each  $\mathcal{E}_i = M$  i.i.d. trained models  $\theta_{i,j}$ .
- Indices:  $I \in \{1, \dots, S\}$  picks a sub-ensemble;  
 $J \in \{1, \dots, M\}$  picks a member.

## Predictive distributions

$$p(y \mid x, \mathcal{E}_i) = \frac{1}{M} \sum_j p(y \mid x, \theta_{i,j})$$

$$p(y \mid x, \mathcal{E}_{1:S}) = \frac{1}{SM} \sum_{i,j} p(y \mid x, \theta_{i,j})$$



## Two MI quantities

- $\mathbb{I}[Y; \mathcal{E}_I \mid x]$ : across sub-ensembles.
- $\mathbb{I}[Y; \theta_{I,J} \mid \mathcal{E}_i, x]$ : within a sub-ensemble.

# Theory: The Main Collapse Result

**Step 1.** Each sub-ensemble is a Monte-Carlo estimator of the BMA:

$$p(y \mid x, \mathcal{E}_i) = \frac{1}{M} \sum_j p(y \mid x, \theta_{i,j}) \xrightarrow{M \rightarrow \infty} \mathbb{E}_{p(\theta)}[p(y \mid x, \theta)] =: p^*(y \mid x).$$

Every sub-ensemble converges to the *same*  $p^*$ , so index  $i$  becomes irrelevant.

**Step 2 (consequence).** If predictions don't depend on  $\mathcal{E}_I$ , then MI between  $Y$  and  $\mathcal{E}_I$  vanishes:

$$\boxed{\mathbb{I}[Y; \mathcal{E}_I \mid x] \longrightarrow 0 \quad \text{as } M \rightarrow \infty} \quad \textit{Epistemic uncertainty collapse.}$$

Growing  $M$  at fixed  $S$  makes the sub-ensembles indistinguishable. The hierarchical ensemble stops being hierarchical from the outside.

# Theory: Across- and Within-Ensemble Disagreement

$$\mathbb{I}[Y; \theta_{I,J} | x] = \mathbb{I}[Y; \mathcal{E}_I | x] + \mathbb{I}[Y; \theta_{I,J} | \mathcal{E}_I, x].$$

## Reading of the chain rule.

- For small sub-ensemble size  $M$ , different sub-ensembles produce distinct predictive distributions, so both terms contribute.
- For large  $M$ , each sub-ensemble approaches the Bayesian model average, and the across-sub-ensemble term  $\mathbb{I}[Y; \mathcal{E}_I | x]$  vanishes.
- The within-sub-ensemble term  $\mathbb{I}[Y; \theta_{I,J} | \mathcal{E}_I, x]$  absorbs the remaining member-level disagreement.

# Theory: A Disagreement Budget

$$\underbrace{\mathbb{I}[Y; \theta_{I,J} \mid x]}_{\text{total disagreement}} = \underbrace{\mathbb{I}[Y; \mathcal{E}_I \mid x]}_{\text{across sub-ensembles}} + \underbrace{\mathbb{I}[Y; \theta_{I,J} \mid \mathcal{E}_I, x]}_{\text{within each sub-ensemble}}$$

**Picture.** Imagine a fixed “budget” of disagreement.

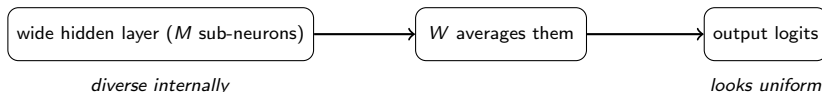
- Small  $M$ : sub-ensembles are noisy and differ from each other, so most of the budget sits *across*.
- Large  $M$ : sub-ensembles each saturate the BMA, the across term empties out, and all budget is spent *within*.

**Practical moral.** Be careful about estimating epistemic uncertainty from already-averaged predictors. Averaging can hide the disagreement we want to measure.

# Proposed Explanation: From Explicit to Implicit Ensembling

**Hypothesis (not proven).** A single wide network may behave internally in a manner analogous to an ensemble of ensembles:

- Matrix multiplication  $Wh$  (weight matrix  $W$  times hidden activation vector  $h$ ) averages across hidden units.
- Global average pooling averages across spatial positions.
- Softmax and logits aggregate many “sub-predictors” into one output.



**Testable prediction.** Under this hypothesis, wider single networks would exhibit an MI collapse similar to that of stacked ensembles; the experimental section evaluates this prediction.

# Theoretical Context: Connection to NTK Theory

Neural Tangent Kernel (NTK) theory (Jacot et al. 2018; Lee et al. 2019):

- In the infinite-width limit, a neural network trained with gradient descent evolves as a linear model in the tangent space of its initialisation.
- All random initialisations converge to the *same* predictor  $f^*$ .

**Consequence for uncertainty.** In an ensemble of infinitely wide nets, every member equals  $f^*$ , so disagreement = 0 and  $MI = 0$ . This is exactly the collapse seen in the ensemble-of-ensembles theory, now for a *single*-model family.

**Two complementary perspectives.**

- *NTK limit*: the predictive function itself converges.
- *Implicit ensembling (hypothesis)*: internal diversity may persist but be averaged away at the output.

The second perspective motivates extracting hidden diversity from a single trained model.

# Empirical Evaluation: Main Questions

The empirical section evaluates three claims suggested by the theoretical analysis and the implicit ensembling hypothesis:

- 1 Does epistemic uncertainty collapse occur in explicit hierarchical ensembles?
- 2 Does model width induce a similar reduction in MI-based uncertainty?
- 3 Can implicit ensemble extraction recover hidden predictive diversity?

## Metrics used throughout

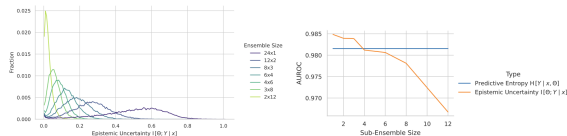
- Predictive entropy  $H$  (total uncertainty, single-model baseline).
- Mutual information  $\mathbb{I}$  (epistemic, primary metric).
- AUROC for OoD detection; classification-with-rejection curves.

# Experiment 1: Explicit Ensembles on CIFAR-10 vs. SVHN

24 independently trained ResNet-50 models, partitioned with  $SM = 24$ :  
( $24 \times 1$ ), ( $12 \times 2$ ), ( $8 \times 3$ ), ( $6 \times 4$ ), ( $4 \times 6$ ), ( $3 \times 8$ ), ( $2 \times 12$ ).

## Observations

- The MI histogram on SVHN collapses toward 0 as  $M$  grows (subfig a).
- AUROC for OoD detection with MI decays slowly:  $\approx 0.988 \rightarrow 0.970$  (subfig b).
- Entropy-based OoD detection degrades less.



(a) Mutual Information Histogram on SVHN (OoD) (b) OoD Detection for CIFAR10 vs. SVHN (OoD)

**Take-away.** Realistic 24-model ResNet ensembles exhibit the predicted collapse. The effect is modest in AUROC but clear in the score distribution.



## Experiment 2: Effect of MLP Width on Mutual Information

Two-hidden-layer MLP, inner layer scaled by various factors; ten-model ensemble per width.  
Representative columns:

| Width                   | $\times 1$ | $\times 4$ | $\times 64$ |
|-------------------------|------------|------------|-------------|
| MNIST accuracy          | 97.97      | 98.59      | 98.63       |
| Mean MI on MNIST        | 0.040      | 0.023      | 0.010       |
| Mean MI on FashionMNIST | 0.228      | 0.192      | 0.114       |

**Interpretation.** Accuracy barely changes, but MI drops sharply. This motivates the collapse interpretation: a wider network does not know less, yet its MI-based uncertainty looks smaller, even on OoD FashionMNIST.

# Experiment 3: Implicit Ensemble Extraction

If a single trained model contains diverse internal predictors that are averaged away at the output, one may attempt to recover this diversity *post hoc* without retraining the backbone.

Two extraction methods are considered:

- (a) **Mask-based extraction on MLPs:** learn boolean masks over the weights of a trained  $\times 64$  MLP to obtain  $K=10$  sub-models.
- (b) **Spatial extraction from pre-trained vision models:** replace global average pooling with adaptive pooling to a  $k \times k$  grid, and treat the  $k^2$  per-tile logits as an ensemble.

## Purpose

These extraction methods test whether a single trained model contains substructures with diverse predictions that are hidden by the final averaged output.

## Experiment 3a: Mask-Based Extraction from an MLP

**Setup.** Fix a width- $\times 64$  MLP (one member of the original ensemble). For each of  $K=10$  sub-models and each layer  $\ell$ , learn a binary mask  $m_\ell^{(k)}$ .

**Objective** (trained only on masks, backbone frozen):

$$\min_m \mathcal{L}_{\text{CE}} - \lambda \cdot \mathbb{I}[\text{sub-model idx; predictions}], \quad \lambda = 2.0.$$

**Results (mean MI):**

|                               | MNIST | Dirty-MNIST | FashionMNIST |
|-------------------------------|-------|-------------|--------------|
| $\times 4$ ensemble           | 0.023 | 0.149       | 0.192        |
| $\times 64$ single            | 0.010 | 0.088       | 0.114        |
| $\times 64$ decomposed (ours) | 0.015 | 0.077       | 0.086        |

**Interpretation.** *Partially* recovers MI on MNIST, but recovery is *not consistent* across Dirty-MNIST and FashionMNIST. This gives evidence that diverse subnetworks can be extracted, but it is not yet a uniformly better uncertainty estimator.

## Experiment 3b: Spatial Extraction from Vision Models

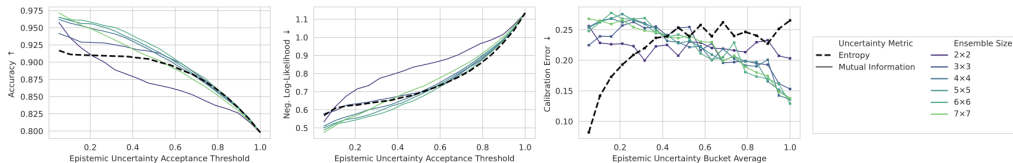
- 1 Replace **global average pooling** with adaptive average pooling to a  $k \times k$  grid.
- 2 For each of the  $k^2$  tiles, read out a logit vector.
- 3 Treat the  $k^2$  vectors as an ensemble; compute MI on them.

Models evaluated on **ImageNet-v2**:

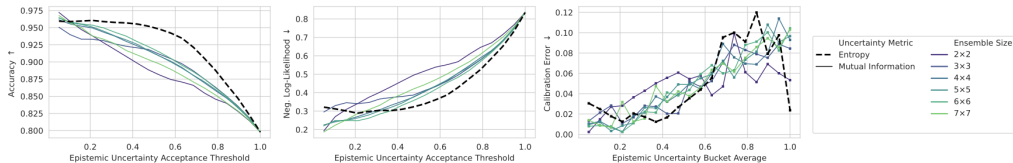
- ResNet-152, Wide-ResNet-101-2, ResNeXt-101-64 $\times$ 4d: extracted ensemble sizes  $k \in \{2, \dots, 7\}$ .
- Compared against the original single model on accuracy, NLL, and calibration error.

*Intuition.* Before global average pooling, each spatial tile carries a slightly different view of the image. Aggregating these per-tile predictions treats them as a small ensemble whose disagreement we can measure.

# Experiment 3b: Classification with Rejection on ImageNet-v2



(a) ResNet models with temperature 1.0



(b) ResNet models with optimal temperature

*Paper Fig. 5(a)–(b): ResNet models with temperature 1.0 (top) and optimal temperature (bottom).*

# Reading the Rejection Curves for ResNet Models

## How to read the figure:

- Samples are sorted by epistemic uncertainty.
- Low-uncertainty samples are accepted first.
- A useful uncertainty score should give higher accuracy and lower NLL at low acceptance thresholds.

## Main message from Fig. 5(a)–(b):

- Without temperature scaling, larger extracted ResNet ensembles improve selective accuracy and NLL, but calibration error is worse.
- With optimal temperature scaling, calibration improves and the original single model benefits most.
- The extracted ensembles are useful for ranking samples by reliability, but calibration remains mixed.

# Additional Evidence: Random Forest Ensembles

**Experiment.** The toy regression is repeated with random forests (max depth 3,  $T \in \{1, 2, 4, 8, 16, 32, 64\}$  trees), forming ensembles of ten forests for each configuration.

**Result.** A similar collapse is observed:

- Forest predictions converge toward the true function and the uncertainty bands narrow.
- The epistemic-uncertainty score decreases by an order of magnitude.
- Mean uncertainty is inversely related to forest size.

**Interpretation.** This suggests that uncertainty collapse is not limited to neural-network optimization, and can arise from hierarchical averaging itself.

# Relation to Other Collapse Phenomena

- **Feature collapse** (Van Amersfoort et al. 2020): intermediate representations become too similar, causing input-space overconfidence. Lives at the *representation* level.
- **Neural collapse** (Papayan et al. 2020): in the terminal phase of training, features and classifier snap to an equiangular tight frame. A *geometric* statement, specific to classification.
- **Epistemic uncertainty collapse** (*this paper*): predictive MI drops as width or hierarchy grows. Lives at the *output distribution* level, and applies to regression too.

## Adjacent extraction ideas.

- *Masksembles* (Durasov 2021): stochastic masks during training.
- *Lottery tickets* (Frankle & Carbin 2018): sparse sub-networks found while training.

*Both are training-time methods; here we extract post hoc from a frozen backbone.*



# Discussion: Limitations

- Implicit ensembling is a hypothesis *consistent with* the observations; the paper does not prove it is the mechanism.
- Mutual information is not always the best uncertainty metric, and the optimal extracted ensemble size  $k$  depends on architecture and temperature.
- On ResNets the extraction *improves accuracy and NLL* but *hurts calibration* in parts of the uncertainty range.
- Experiments focus on vision; no language or multimodal evidence.

# Main Takeaways

- 1 **Lower MI is ambiguous.** It may reflect better learning, but it may also reflect averaged-away disagreement.
- 2 **Averaging matters.** Hierarchical or implicit averaging can hide the variation that MI-based uncertainty estimators rely on.
- 3 **Extraction is diagnostic.** Mask-based or spatial extraction can reveal hidden predictive diversity, but is not yet a universal solution.
- 4 **The mechanism remains open.** Implicit ensembling is supported by the evidence, but not proved as the unique cause.

# References and Thanks

- Kirsch, A. *(Implicit) Ensembles of Ensembles: Epistemic Uncertainty Collapse in Large Models*. TMLR, 05/2025.
- Fellaji, M., Pennerath, F. *The Epistemic Uncertainty Hole: an issue of Bayesian Neural Networks*. arXiv:2407.01985, 2024.
- Smith, L., Gal, Y. *Understanding measures of uncertainty for adversarial example detection*. arXiv:1803.08533, 2018.
- Lakshminarayanan, B. et al. *Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles*. NeurIPS 2017.
- Gal, Y. et al. *Deep Bayesian Active Learning with Image Data*. ICML 2017.
- Jacot, A. et al. *Neural Tangent Kernel*. NeurIPS 2018. Lee, J. et al. NeurIPS 2019.
- Van Amersfoort, J. et al. *Uncertainty Estimation Using a Single Deterministic Model*. ICML 2020.
- Papayan, V., Han, X., Donoho, D. *Prevalence of neural collapse during the terminal phase of deep learning training*. PNAS 2020.
- Durasov, N. et al. *Masksembles*. CVPR 2021. Frankle, J., Carbin, M. ICLR 2019.

Thank you! Questions?